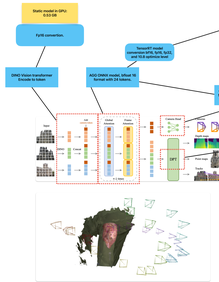
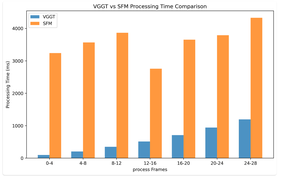




CPU requirements:

AliceVision need 10 threads
VGGT only need 1 thread(4 threas for better performance)
Num of image requirements:

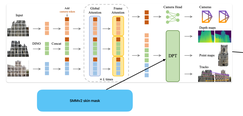
AliceVision need high overlap, 30 images for reconstruction.
VGGT can work with more than 1 images
GPU memory requirements:

AliceVision need 3GB GPU memory.(Feature detection and matching)
VGGT need 5.4GB (half precision on RTX 4090 with tensorRT dino vision transformer, attention layer model with 2 heads), (19GB for float 32 pytorch)
Cloud point density.

same as all image resolution, N * 518 * 518 points.
only feature points. (very low)

TSDF Integration

SKANet Integration



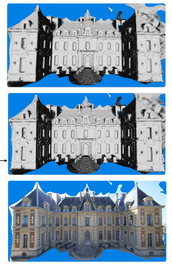
Mesh Reconstruction

This point cloud is not good for mesh reconstruction

TSDF

Single layer mesh but down sampled, good for mesh reconstruction, but not good for high resolution 3D image generation for segmentation

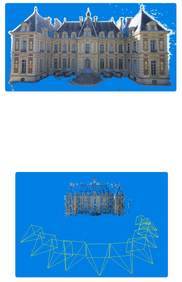
OpenMVS



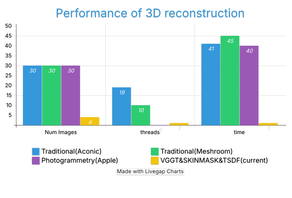
Raw mesh reconstruction

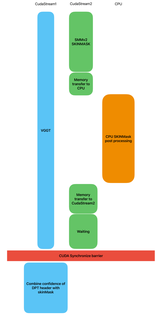
Mesh refinement.

Mesh Texturing

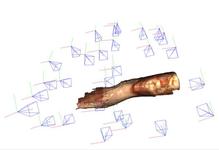


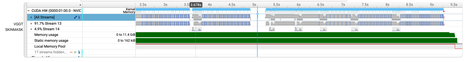
4 frame: 114ms
4-8 frame: 222ms
8-12 frame: 358 ms
12-16 frame: 519 ms
16-20 frame: 702ms
20-24 frame: 919ms
24-28 frame: 1160ms



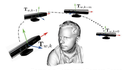


SKIN MASKED VGGT works successfully



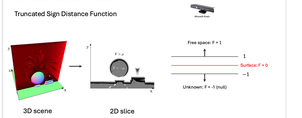


KinectFusion: Background



KinectFusion: TSDF

Truncated Sign Distance Function



KinectFusion: TSDF

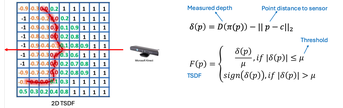
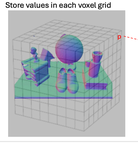


Image credit: https://tinext.io/understanding-real-time-3d-reconstruction-and-kinectfusion-33d651d1cd402

KinectFusion: TSDF

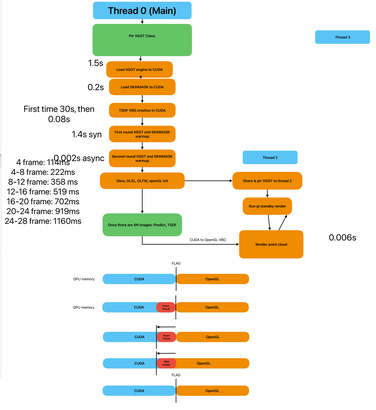
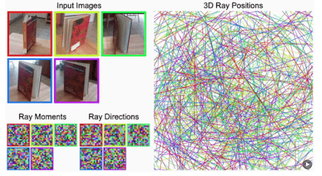
Store values in each voxel grid



Weight is high if the sensor is near and perpendicular

KinectFusion: Volume Integration

Each sensor provides a partial (noisy) observation of the signed distance function

Volume integration recursively update the map by integrating new observation.

Incremental point cloud inferencing result and speed